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REVIEW ARTICLE

Social isolation and social connectedness among young adult cancer survivors: A systematic review

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Abstract

Background: Social isolation and connectedness are social determinants of health that have demonstrated effects on cancer-related outcomes. These constructs have been systematically evaluated among pediatric and older adult cancer populations. In this review, the authors evaluated the prevalence, correlates, and psychosocial implications of social isolation and connectedness among young adult (YA) cancer survivors aged 18–39 years.

Methods: Peer-reviewed articles published in English before June 2021 were identified from database searches and included articles' reference lists according to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. Included articles described studies that assessed social isolation and/or connectedness among YA cancer survivors.

Results: In total, 5094 unique records were identified; 4143 were excluded after title/abstract screening, and 907 were excluded after full-text review. Forty-four articles were included. Few studies used validated measures or directly assessed social isolation or connectedness. Social isolation was similarly prevalent among YAs and older cancer survivors and noncancer populations. Demographic, clinical, and behavioral risk and protective factors for social isolation were identified. Social isolation was related to worse psychological well-being, whereas social connectedness was often, but not always, related to better psychological well-being.

Conclusions: This growing literature underscores the relevance of social isolation and connectedness as important health determinants among YA cancer survivors. The identified risk and protective factors can identify YAs who especially may benefit from screening for social isolation. Future studies are needed that directly,

The last two authors contributed equally to this work.

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reliably, and validly evaluate social isolation and connectedness to inform the development of interventions to decrease isolation and increase connectedness.

KEYWORDS
cancer survivorship, social connectedness, social isolation, systematic review, young adults

More than 80,000 young adults (YAs) aged 18–39 years are diagnosed with cancer in the United States annually.[1] YAs with cancer face disease-related and treatment-related life disruptions that can negatively affect well-being.[2–4] In developed countries, young adulthood is often when individuals complete formal education, launch careers, begin living independently, achieve financial independence, establish long-term emotional, romantic, and sexual relationships, and start families.[5,6] Core social relationships shift from being primarily with parents to including friends, colleagues, partners, and perhaps children.[5] However, YAs who have been diagnosed with cancer can experience delays in these sociodevelopmental milestones. Thus the implications of cancer for social well-being may be particularly significant for YAs.

The construct of social well-being can be viewed through two lenses. According to Holt-Lunstad and colleagues,[7] *social isolation* is a lack of interaction, contact, and/or relationships with other individuals and/or with society at large,[8,9] whereas *social connection* comprises the combined structural and functional components of social relationships.[7] Broadly conceptualized as umbrella terms, social isolation and connectedness encompass the ways in which individuals connect with others through various channels (e.g., emotional, behavioral, physical).[7] Therefore, several related constructs are conceptualized within social isolation (e.g., loneliness) and social connectedness (e.g., social support).

Social isolation is an important and well established determinant of health that is consistently linked to all-cause mortality, morbidity, and negative mental health.[10] In cancer, it is associated with low treatment engagement and mortality.[11] Although most research on social isolation has focused on older adults, emerging evidence underscores its relevance for YAs. For example, loneliness is more prevalent among YAs than older adults,[12,13] and YAs with long-standing illness, such as cancer, have higher incidence and severity of loneliness than healthy peers.[14]

Social connectedness is another determinant of health that is theorized to mechanistically underlie the health-promoting effects of social relationships.[15] It is linked to longer life expectancy, better mental health, better cognitive functioning, and enhanced neuroendocrine and immune regulation.[15] In cancer, social connectedness is associated with improved social and emotional well-being,[16] and increased social support is associated with decreased mortality.[17] Thus social connectedness and aspects thereof, such as social support and belongingness, could promote well-being for YAs with cancer.

To date, two known systematic reviews have focused on social well-being among YA cancer survivors (i.e., YAs from the point of cancer diagnosis through the end of life[18]). Warner and colleagues[19] reported that cancer survivors aged 15–39 years reported difficulties related to employment, educational attainment, financial stability, interpersonal relationships, and peer support. Although numerous areas of need were identified, the review did not address the prevalence and unique impact of social isolation or connectedness among YA cancer survivors. Pahl and colleagues[20] conducted a systematic review of social isolation and connectedness among cancer survivors aged 10–21 years who were diagnosed during childhood or adolescence. However, the normative sociodevelopmental transitions and cancer-related challenges faced by this younger cohort are different from those faced by YAs. Therefore, the prevalence and correlates of social isolation and connectedness in YA cancer survivors remain unclear.

To address this gap, we systematically evaluated the literature regarding social isolation and social connectedness in YA cancer survivors aged 18–39 years. Within this population, our objective was to determine: (1) the prevalence of social isolation and connectedness (i.e., how commonly they are experienced), (2) clinical and sociodemographic correlates of social isolation and connectedness, and (3) relationships of social isolation and connectedness with psychological well-being.

METHODS

We searched MEDLINE (National Library of Medicine, National Institutes of Health), the Cochrane Database of Systematic Reviews, PROSPERO (an international prospective register of systematic reviews), and the Joanna Briggs Institute Systematic Review Register and confirmed that there were no other systematic reviews on this topic. The protocol was registered on PROSPERO in advance (CRD42021282885).[21] The methodology and results are reported according to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses)[22] guidelines.

Inclusion criteria

Eligible articles described empirical studies published in English that assessed social isolation and/or connectedness among YA cancer survivors who were aged 18–39 years at the time of diagnosis and 18–44 years at the time of participation. Multiple articles describing the same sample were retained if each presented different relevant results. The age range for participation was extended beyond 39

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years to ensure that longitudinal studies of YA experiences were included. We focused on YAs aged 18 years and older rather than 15 years or older to minimize the likelihood that participants transitioned from a pediatric to an adult treatment setting during the study. Articles with participants outside this age range were included if they presented results for participants aged 18–39 years or a subset thereof separately. Articles that did not report participants' ages at diagnosis were included if they clearly described YA cancer survivors' experiences. Consistent with Pahl and colleagues,[20] we operationalized social isolation and social connectedness according to Holt-Lunstad and colleagues'[7] framework. Therefore, studies were included if they assessed social isolation, social connectedness, or a related concept (e.g., loneliness, social support).

Search strategy

Articles were identified by searching MEDLINE (Ovid), PsycINFO (EBSCO Information Services), EMBASE (Excerpta Medica Database; Elsevier), and CINAHL (Cumulative Index of Nursing and Allied Health; EBSCO Information Services) in June 2021. A medical librarian developed a search string for EMBASE and subsequently modified it for the remaining systems (see Supporting Methods). Reference lists of included articles were also reviewed. All study designs were included; however, gray literature was excluded. No restrictions were imposed based on publication date, research setting, cancer type, or treatment status.

Study selection

Identified article citations were uploaded into the Rayyan digital platform,[23] and duplicates were removed. To calibrate the screening protocol, three reviewers (G.E.A., T.F.D.V., J.S.G.) screened the titles and abstracts of the same 77 articles. The reviewers then met as a group with R.S.F. and L.B.O. to resolve conflicts. Reviewer pairs screened the remaining titles and abstracts. The full texts of articles that were not excluded were retrieved and screened. Screening conflicts were resolved through consensus review and discussion.

Assessment of methodological quality

We assessed study quality with the *Standard Quality Assessment Criteria for Evaluating Primary Research Papers* (QualSyst),[24] which is appropriate for quantitative (14 criteria) and qualitative (10 criteria) designs. For each study, relevant criteria were scored 0 (not met), 1 (partially met), 2 (met), or not applicable. Criteria scores were summed up and divided by the total possible score based on the number of applicable items. Higher scores represented better methodological quality. Scores were categorized as *limited* (<0.50), *adequate* (0.50–0.70), *good* (0.71–0.79), or *strong* (≥0.80) quality.[25,26]

Data extraction

Data were extracted and recorded in a standardized Excel form (Microsoft Corporation). In addition to the extraction plan outlined in the preregistered protocol, we documented the methods used to assess psychological well-being when available. We defined *psychological well-being* as any aspect of emotional or affective health; however, we excluded broad conceptualizations of health-related quality of life because these can include components of physical and social health.[27] All reviewers pilot-tested the extraction form with six articles and met as a group with R.S.F. and L.B.O. to resolve conflicts. Reviewer pairs then extracted data from the remaining articles. Conflicts were resolved through consensus review and discussion. For studies that included broader samples, data were only extracted for participants who were aged 18–44 years at participation.

RESULTS

Search results

As shown in Figure 1, the initial database search yielded 7414 titles. After de-duplication, 5050 titles and abstracts were screened. Of these, 4124 were excluded, and 926 full texts were retrieved. Of the full texts reviewed, 887 were excluded, and 39 were retained. A search of the included articles' reference lists identified an additional 44 articles for screening. Of these, 19 were excluded based on the title and/or abstract, and 20 were excluded after a full-text review, yielding five additional articles. In total, 44 articles were included.

Study characteristics

As outlined in Table 1, most studies used qualitative methods (61.4%), and nearly all were observational (97.7%). Studies were predominantly conducted in the United States (45.5%) or Canada (15.9%), included participants with mixed cancers (72.7%) or breast cancer (20.5%), included participants who were receiving active anticancer treatment (36.0%), and were published since 2010 (81.8%). Table S1 (quantitative studies) and Table S2 (qualitative studies) show the results of the methodological quality assessments. Overall study quality was strong, with QualSyst scores ranging from 0.75 to 1.00 (mean, 0.91) for quantitative studies and from 0.65 to 1.00 (mean, 0.85) for qualitative studies.

Prevalence of social isolation and social connectedness among YA cancer survivors

Most articles (n = 37; 86.4%) reported the prevalence of social isolation and connectedness (Table 2). Within those articles, 11 (29.7%) used quantitative methods, 25 (67.6%) used qualitative methods, and one (2.7%) used mixed methods. Quantitative studies

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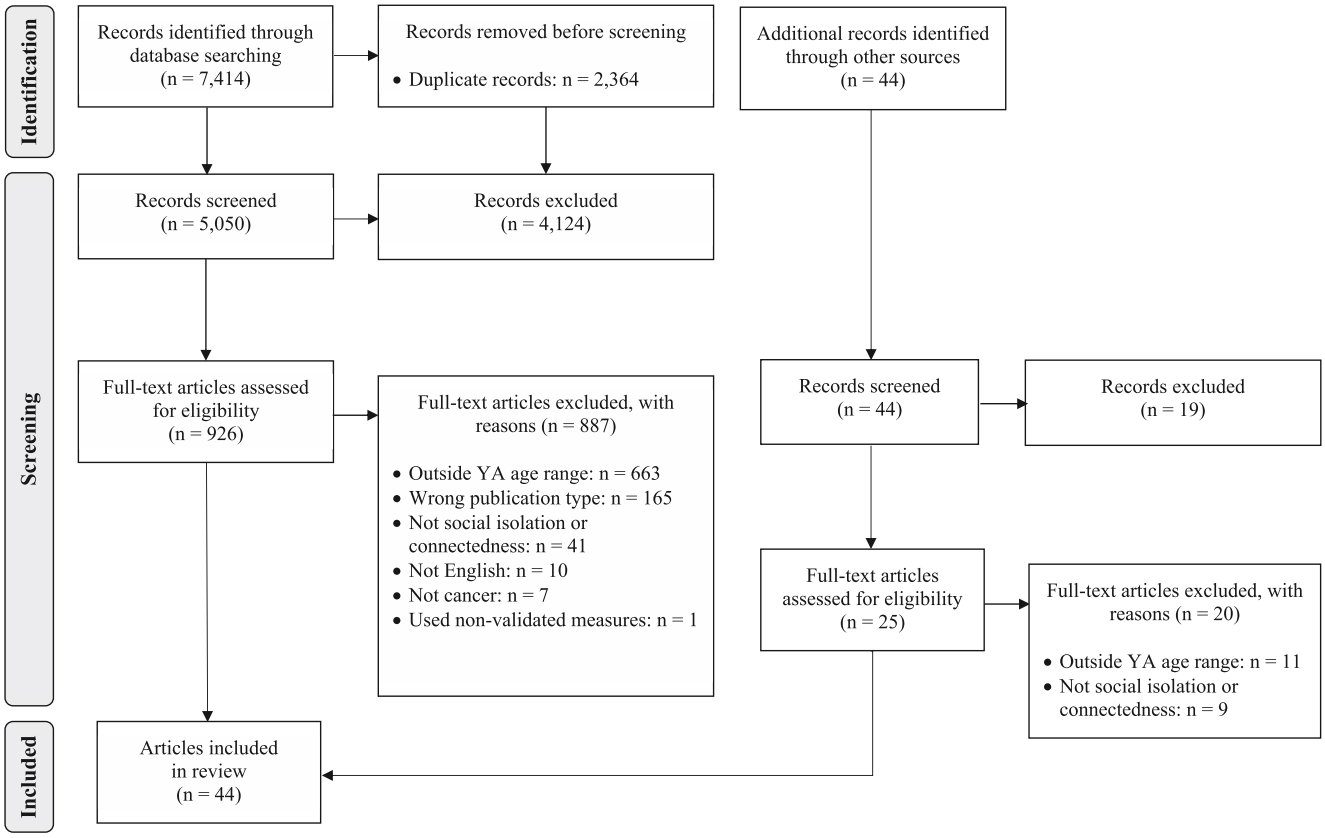

FIGURE 1 PRISMA flowchart. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; YA, young adult.

indicated that, relative to older cancer survivors and cancer-free individuals, YAs reported similar levels of social isolation/connectedness and related constructs (e.g., loneliness[28] and social support[29–31]). For some YAs, cancer positively affected their relationships with family and friends[32] and increased the value they placed on those relationships.[33] In qualitative studies, YAs identified family, friends, neighbors, colleagues, other cancer survivors, and health care professionals as key sources of connectedness.[34–36]

Conversely, some studies indicated that YAs had elevated levels of social isolation. For example, Paskett and colleagues[37] observed that YA breast cancer survivors reported worse social functioning than cancer-free peers and older survivors. Similarly, quantitative studies indicated that, relative to other populations, YAs expressed more detrimental interactions,[30] diminished social support,[38,39] and lower social functioning.[28] Themes related to social isolation emerged in numerous qualitative studies. Some focused on topics closely related to isolation and connectedness.[36,40–46] However, others focused on more distant concepts, such as coping,[47] uncertainty,[34,48,49] fertility,[50] physical activity, nutrition, weight,[51] and the general lived experience YA cancer survivors.[52–59] In some studies, isolation was directly related to the disease experience. For example, Halliday and colleagues[50] observed that young female hematologic cancer survivors described a sense of "otherness" associated with having uncertain fertility. Kenen and colleagues[42] observed that social separation was exacerbated by carrying a BRCA mutation. Knox and colleagues[52] identified advanced cancer as particularly isolating because YAs reported feeling poorly understood by both cancer-free peers and YAs with localized disease.

Two studies identified the transition off treatment as particularly isolating,[60,61] and four identified isolation from other YA cancer survivors as a major concern.[44,54,56,61] Conversely, in three studies, YAs who connected with peers experienced greater social connectedness.[53,59,62] Kaal and colleagues[62] reported lower social isolation and greater connectedness among YAs engaged with an online YA support community. Kim and Gillham[53] observed that online communities helped YAs befriend other YA cancer survivors, increasing their sense of connectedness. Finally, Hauken and colleagues[59] observed that YAs who met other YA cancer survivors expressed greater feelings of connectedness.

Clinical and sociodemographic correlates of social isolation and social connectedness

Clinical and sociodemographic correlates of social isolation and connectedness were identified in 18 articles (40.9%), although few correlates were explored in more than one study (Table 3). Within these articles, six (33.3%) used quantitative methods, 11 (61.1%) used qualitative methods, and one (5.6%) used mixed methods. Quantitative studies indicated that YAs reporting greater social support were more likely to be physically active,[38] White, female, married, and diagnosed with breast (vs. another) cancer,[63] employed

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TABLE 1 Characteristics of young adult participants in the included articles.

Reference	Design	Country	No. of YAs	Age at Dx: Mean ± SD [range], years	Age at study: Mean ± SD [range], years	Months since Dx	Tx status: Time since Tx ended	Race/ethnicity, sex	Cancer type(s)	Assessment of social isolation/social connectedness	Assessment of psychological well-being
Quantitative studies											
Avis 2012[29]	Obs	USA	132	NR [NR]	NR [25–44]	≤8	On	88% NHW, NR, 100% female	Breast	MOS Social Support Survey	Brief Depression Inventory
Bellizzi 2012[32]	Obs	USA	440	NR [21–39]	NR [NR]	<14	Mixed	59% NHW, 21% H/L, 63% female	Mixed	AYA Hope Patient Survey	NA
Brunet 2014[38]	Obs	NR	64	NR [NR]	28.8 ± 5.5 [20–39]	Mean, 34.8	Posttreatment: Mean, 2.9 years	95% White, NR, 73% female	Mixed	Social Provisions Scale	Perceived Stress Scale-10
Geue 2019[70]	Obs	Germany	514	29.6 ± 6.14 [18–39]	<4 years postdiagnosis	≤48	On	NR, NR, 75% female	Mixed	Illness-specific Social Support Scale-8	Hospital Anxiety and Depression Scale Single item assessing overall life satisfaction
Geue 2019[30]	Obs	Germany	179	NR [18–39]	27.5 ± 5.79 [19–42]	12.1	On	NR, NR, 62% female	Hem	Illness-specific Social Support Scale-8	European Organization for the Research and Treatment of Cancer Quality-of-Life Questionnaire-Core 30
Hoyt 2013[71]	Obs	USA	284	NR [NR]	25.2 ± 3.32 [18–29]	32.4	On	46% NHW, 38% H/L, 0% female	Testicular	Cancer Assessment for Young Adults-Testicular	Functional Assessment of Cancer Therapy-General European Organization for the Research and Treatment of Cancer Quality-of-Life Questionnaire-Core 30 Cancer Assessment for Young Adults-Testicular
Kaal 2018[62]	Obs	Netherlands	30	25.6 ± 6.4 [18–35]	29.8 ± 5.3 [22–39]	2.7	On	NR, NR, NR	Mixed	13 items assessing the usefulness of an online AYA community	NA
Naik 2020[28]	Obs	Canada	420	NR [18–39]	NR [18–39]	≤6	On	NR, NR, 100% female	Breast	Canadian Problem Checklist	NA

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TABLE 1 (Continued)

Reference	Design	Country	No. of YAs	Age at Dx: Mean ± SD [range], years	Age at study: Mean ± SD [range], years	Months since Dx	Tx status: Time since Tx ended	Race/ethnicity, sex	Cancer type(s)	Assessment of social isolation/social connectedness	Assessment of psychological well-being
Okamura 2021[69]	Obs	Japan	206	NR [NR]	33.7 ± 4.3 [22–39]	<1 to ≥120	Mixed	NR, NR, 87% female	Mixed	Multidimensional Scale of Perceived Social Support-short form	Supportive Care Needs Survey
Paskett 2008[37]	Obs	USA	11	NR [NR]	NR [25–44]	Median, 145.2	Posttreatment, NR	NR, NR, 100% female	Breast	MOS Social Support Survey MOS SF-36 (social functioning scale)	NA
Roper 2013[33]	Obs	USA	40	NR [NR]	30.41 ± 5.7 [21–40]	Newly dx	Posttreatment, 1–6 months	90% NHW, NR, 60% female	Lymphoma	Impact of Cancer (social and relationship scale)	NA
Trevino 2013[63]	Obs	USA	71	NR [NR]	34.0 ± 5.57 [20–40]	Mean, 40.8	On	93% White, 2% H/L, 70% female	Mixed	Interpersonal Support Evaluation List-12	McGill Quality-of-Life Questionnaire (psychology and existential subscales)
Trevino 2013[39]	Obs	USA	95	NR [NR]	33.4 ± 5.51 [20–40]	Mean, 43.44	On	86% White, 5% H/L, 68% female	Mixed	Interpersonal Support Evaluation List-12	NA
Xie 2017[31]	Obs	China	524	NR [NR]	NR [21–39]	NR	NR	NR, NR, 60% female	Mixed	Social Support Rating Scale	Distress Thermometer Hospital Anxiety and Depression Scale
Zebrack 2017[64]	Int	USA	304	NR [18–40]	NR [18–40]	NR	Mixed	79% White, NR, 82% female	Mixed	Duke/UNC Functional Social Support Scale	Patient Health Questionnaire-4
Qualitative studies											
Corbeil 2009[34]	Obs	Canada	6	NR [NR]	25 [19–30]	Mean, 17.5	On	NR, NR, 50% female	Mixed	Semistructured interviews	NA
D'Agostino and Edelstein 2013[40]	Obs	Canada	9	24.7 ± 3.97 [18–29][a]	30.0 ± 3.16 [23–34][a]	NR	Posttreatment, ≥2 years	NR, NR, 56% female	Mixed	Focus groups	NA
Easley 2013[45]	Obs	Canada	12	32 ± 3.94 [24–37]	34 ± 3.26 [28–38]	NR	Posttreatment, 1–5 years	100% White, NR, 92% female	Thyroid	Interviews	Interviews

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TABLE 1 (Continued)

Reference	Design	Country	No. of YAs	Age at Dx: Mean ± SD [range], years	Age at study: Mean ± SD [range], years	Months since Dx	Tx status: Time since Tx ended	Race/ethnicity, sex	Cancer type(s)	Assessment of social isolation/social connectedness	Assessment of psychological well-being
Halliday 2014[50]	Obs	Australia	12	NR [NR]	NR [25–39]	≥12	Posttreatment, NR	NR, NR, 100% female	Hem	Interviews	NA
Hamid and Khan 2021[36]	Obs	India	10	31.9 ± 6.31 [20–38][a]	28.6 ± 6.43 [22–41][a]	≤6	On	NR, NR, 100% female	Breast	Semistructured interviews	Semistructured interviews
Hanghøj 2021[66]	Obs	Denmark	13	NR [NR]	23.5 [18–29]	NR	Mixed	NR, NR, 69% female	Mixed	Semistructured interviews	Semistructured interviews
Hauken 2013[59]	Obs	Norway	20	NR [NR]	31.1 ± 3.9 [24–35]	Mean, 24.6	Posttreatment: Mean, 16 years	NR, NR, 75% female	Mixed	Semistructured interviews	NA
Hauken and Larsen 2019[41]	Obs	Norway	20	NR [NR]	31.1 ± 3.9 [24–35]	Mean, 24.6	Posttreatment: Mean, 16 years	NR, NR, 75% female	Mixed	Semistructured interviews	NA
Kenen 2006[42]	Obs	UK	10	NR [NR]	41.4 ± 1.51 [39–44][a]	NR	On	NR, NR, 100% female	Breast, ovarian	Focus groups	Focus groups
Kim and Gillham 2013[53]	Obs	Australia	46	NR [NR]	31 [23–38]	NR	On	NR, NR, 74% female	Mixed	Online blogs	NA
Knox 2017[52]	Obs	Canada	10	NR [NR]	26 ± 10.6 [18–35]	NR	Mixed	NR, NR, 70% female	Mixed	Semistructured interviews	NA
Kumar and Schapira 2013[35]	Obs	USA	15	NR [NR]	NR [18–30]	NR	Mixed, ≤5 years	87% White, 13% H/L, 67% female	Mixed	Semistructured interviews	Semistructured interviews
Lidington 2021[54]	Obs	UK	65	31.7 ± NR [25–39]	33.6 [25–42]	Mean, 22.8	Mixed	75% White, NR, 60% female	Mixed	Semistructured interviews, focus groups	NA
Love 2014[43]	Obs	USA	9	NR [NR]	29.3 [21–39][a]	Mean, 85.2	Mixed	89% White, NR, 0% female	Mixed	Online forum, focus groups, semistructured interviews	NA

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Reference	Design	Country	No. of YAs	Age at Dx: Mean ± SD [range], years	Age at study: Mean ± SD [range], years	Months since Dx	Tx status: Time since Tx ended	Race/ ethnicity, sex	Cancer type(s)	Assessment of social isolation/social connectedness	Assessment of psychological well-being
Miedema 2007[47]	Obs	Canada	13	27.5 ± 4.7 [20–35][a]	30.2 ± 5.3 [20–37][a]	NR	Mixed	NR, NR, 60% female	Mixed	Interviews	NA
Mikkelsen 2008[60]	Obs	Denmark	NR	NR [NR]	NR [NR]	NR	On	NR, NR, NR	Mixed	Focus groups	NA
Miller 2015[48]	Obs	USA	25	34 ± NR [26–39]	NR [NR]	NR	On	100% White, NR, 100% female	Breast	Interviews	NA
Milosevic 2020[51]	Obs	Canada	12	34 ± 3.5 [27–39]	36 ± 3.4 [31–41]	≤60	NR	73% White, NR, 100% female	Breast	Semistructured interviews	NA
Mishra 2018[55]	Obs	USA	8	NR [NR]	31.9 ± 5.2 [23–39][a]	≥6	On	100% White, 75% H/L, 88% female	Mixed	Semistructured interviews	Semistructured interviews
Musiello 2014[56]	Obs	Australia	7	NR [NR][b]	19.1 ± 1.6 [18–22][a]	NR	Mixed, ≤6 months	NR, NR, 71% female	Mixed	Focus groups	Focus groups
Nizamli 2011[57]	Obs	Syria	17	NR [NR]	37 ± NR [30–45][c]	NR	NR	NR, NR, 100% female	Breast	Semistructured interviews	NA
Panjwani 2019[49]	Obs	USA	53	25.9 ± 5.3 [NR]	28.4 ± 5.0 [20–39]	Mean, 28	Mixed	79% White, 8% H/L, 68% female	Hem	Semistructured interviews	NA
Ruddy 2013[61]	Obs	USA	36	NR [NR]	37.8 ± 4.7 [26–44]	Mean, 22.3	On	94% NHW, NR, 100% female	Breast	Focus groups	NA
Siegel 1999[67]	Obs	USA	34	30.6 ± NR [22–35]	NR [NR]	Mean, 38	Posttreatment, ≥6 months	85% White, 0% H/L, 100% female	Breast	Semistructured interviews	NA
Snyder and Pearse 2010[44]	Obs	USA	70	32.9 ± NR [23–39]	35.2 ± NR [NR]	85.7% of sample: <36	Mixed	61% NHW, 4% H/L, 100% female	Breast	Semistructured interviews	Semistructured interviews

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TABLE 1 (Continued)

Reference	Design	Country	No. of YAs	Age at Dx: Mean ± SD [range], years	Age at study: Mean ± SD [range], years	Months since Dx	Tx status: Time since Tx ended	Race/ethnicity, sex	Cancer type(s)	Assessment of social isolation/social connectedness	Assessment of psychological well-being
Victorson 2019[58]	Obs	USA	16	NR [NR]	33.1 ± 3.7 [28–39]	NR	Posttreatment, ≤5 years	50% White, 6% H/L, 75% female	Mixed	Focus groups	NA
Zebrack 2010[65]	Obs	USA	17	NR [NR]	NR [18–35]	NR	Posttreatment NR	NR, NR, NR	Mixed	Focus groups	NA
Mixed-methods studies											
Darabos 2022[68]	Obs	USA	59	NR [NR]	35.1 ± 4.84 [24–42]	<60	Mixed	93% White, NR, 97% female	Mixed	PROMIS Social Isolation eight-item short form Two open-ended questions about social support preferences	NA
Munoz 2016[46]	Obs	USA	31	NR [NR]	33.2 ± 5.1 [21–39]	NR	Posttreatment, ≤5 years	29% Black, 42% H/L, 65% female	Mixed	Semistructured interviews	NA

Note: Participants in the 2019 study by Geue et al.[30] comprised a subset of those in their other 2019 study.[70] Participants in the 2013 study by Trevino et al.[63] were a subset of those in their other 2013 study.[39] Participants in Hauken and Larsen 2019[41] were redundant with those in the 2013 study by Hauken et al.[59]

Abbreviations: AYA, adolescent and young adult; Dx, diagnosis/diagnosed; Hem, hematologic; H/L, Hispanic/Latinox; Int, interventional; MOS, Medical Outcomes Study; NA, not applicable; NR, not reported; Obs, observational; PROMIS, Patient-Reported Outcomes Measurement Information System; SD, standard deviation; Tx, treatment; UNC, University of North Carolina.

[a]Calculated by the review authors from participant-level data provided in the article.

[b]All participants had been diagnosed within the prior 12 months and thus were no more than 1 year older than their age at diagnosis.

[c]No quotes from participants older than 44 years were provided in the article.

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TABLE 2 Results related to the prevalence of social isolation and connectedness.

Reference	Results	Meta-inferences
Quantitative studies		
Avis 2012[29]	YAs reported average levels of social support on the RAND Social Support Scale (mean ± SD, 4.3 ± 0.7, range; 1–5) relative to older adults with cancer.	• In general, YAs reported average-to-elevated levels of social isolation, although there is also evidence that cancer can positively affect important relationships. • Engagement with supportive interventions, such as online communities, was related to decreased reports of social isolation and increased social connectedness. • Few quantitative studies assessed social isolation and/or connectedness directly (i.e., with a baseline measure of social isolation/social connectedness), and often assessed prevalence using indirect means (i.e., social problems, social support, general questions about social relationships).
Bellizzi 2012[32]	On average, 11% of YAs reported that cancer negatively affected their relationships with family and friends, whereas 63% reported that cancer had a positive effect on these relationships.	
Brunet 2014[38]	YAs reported slightly diminished levels of perceived social support on the Social Provisions Scale (mean ± SD, 3.33 ± 0.47; range, 1–4) relative to a sample of college students, public school teachers, and military nurses (Sherbourne & Stewart 1991[88]). One half of respondents reported they had previously been involved in a support group designed for people with cancer.	
Geue 2019[30]	YAs reported elevated levels of detrimental interactions (mean ± SD, 4.2 ± 3.1; range, 0–16) and comparable levels of positive support (mean ± SD, 13.4 ± 2.8; range, 0–16), as measured by the Illness-Specific Social Support Scale, relative to older adult participants. Only 12.3% (n = 22) of YAs reported having no detrimental interactions, and 29.1% (n = 52) had positive support scores at the maximum value of 16.	
Kaal 2018[62]	After engaging with an online YA support community, 13 YAs (48%) reported not feeling lonely anymore, 17 (63%) reported having good contact with peers, 12 (44%) reported they had made new friends, and 17 (63%) reported they felt supported.	
Naik 2020[28]	On the Canadian Problem Checklist, 10.5% of YAs (n = 44) reported feeling alone, comparable to older adults with cancer.	
Paskett 2008[37]	YAs aged 35–44 years reported lower social functioning, as measured by the MOS SF-36 scale (mean ± SD, 73.75 ± 28.53) than the general US female population within the same age group (mean ± SD, 83.07 ± 23.27), and relative to older breast cancer survivors.	
Roper 2013[33]	At the conclusion of treatment through 6 months posttreatment, YAs reported placing higher value on relationships with family and friends than before they were diagnosed with cancer (mean ± SD, 3.9 ± 4.1; range, 1–5) as well as feeling a special bond with people with cancer (mean ± SD, 3.6 ± 4.0; range, 1–5).	
Trevino 2013[39]	Relative to patients with arthritis or at risk for cardiovascular disease and healthy adults (Cohen 2008[86]), YAs reported decreased availability of overall social resources (mean ± SD, 42.38 ± 4.39; range, 12–48) as well as a diminished sense of having someone available to talk to about problems (mean ± SD, 13.97 ± 1.89; range, 4–16), decreased tangible or material aid (mean ± SD, 14.57 ± 1.66; range, 4–16), and a decreased sense of belonging or of having someone with whom to engage in activity (mean ± SD, 13.85 ± 2.11; range, 4–16) as measured by 12 items from the Interpersonal Support Evaluation List.	
Xie 2017[31]	YAs reported moderate levels of social support according to established cutoff points as measured by the Chinese version of the Social Support Rating Scale (mean ± SD, 40.58 ± 6.96; range, 12–66).	

(Continues)

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TABLE 2 (Continued)

Reference	Results	Meta-inferences
Zebrack 2017[64]	YAs reported average levels of social support as measured by the Duke-UNC Functional Social Support Questionnaire (mean ± SD, 4.06 ± 0.81; range, 1–5) relative to older patients with advanced cancer (Applebaum 2014[87]).	
Qualitative studies		
Corbeil 2009[34]	Nearly all YAs reported that they relied heavily on family for support throughout their cancer experience.	• Social isolation-related themes emerged in multiple studies. Some of these studies focused on topics closely related to isolation, such as social support, whereas others targeted more distant concepts, such as fertility, identity formation, physical activity, nutrition, weight management, and the overall experience of being a YA with cancer. • YAs frequently described feeling different from peers, lacking support from peers and family, and feeling concerned about social relationships. • Multiple studies identified feeling isolated from other YAs experiencing cancer as a major concern. • Among those who reported feeling socially connected, important sources of support included family, friends, neighbors, colleagues, other cancer survivors, and health care professionals.
D'Agostino and Edelstein 2013[40]	YAs with cancer expressed concerns about relationships with peers and family, including feelings of social isolation.	
Easley 2013[45]	YAs with thyroid cancer expressed feeling isolated in three ways: (1) from other cancer survivors because of having a "good" cancer (e.g., high chance of survival, no chemotherapy), (2) because of mandatory physical isolation periods after radioactive iodine treatments, and (3) from peers without cancer.	
Halliday 2014[50]	YAs with uncertain fertility described feeling different and disconnected from peers with no cancer history.	
Hamid and Khan 2021[36]	Kashmiri YAs with breast cancer in the study reported receiving support from parents, siblings, and other family members. Other common sources of support included friends, neighbors, colleagues, other women with breast cancer, and health care professionals.	
Hauken and Larsen 2019[41]	Most YAs reported lacking peer support during cancer treatment.	
Hauken 2013[59]	Upon re-entry into everyday life after cancer treatment, YAs reported increased social isolation and a lack of social connectedness with families and friends, employers, colleagues, and health care providers, even when surrounded by family and friends and enrolled in regular follow-ups. Conversely, those who had met other YAs reported feeling greater social connectedness.	
Kenen 2006[42]	YAs with breast cancer carrying a BRCA1 or BRCA2 mutation reported feeling isolated, separate, and alone and identified instances of social separation initiated both by themselves and by others, even in the presence of social support.	
Kim and Gillham 2013[53]	Isolation was one of the primary themes identified in an analysis of online blogs written by YAs. Conversely, engaging with a social network on the internet promoted feelings of social connectedness from both new and existing relationships.	
Knox 2017[52]	An advanced cancer diagnosis universally was experienced as isolating and unexpected. All YAs felt that they did not belong and that others could not understand them, particularly peers with no cancer history or with localized cancer.	
Kumar and Schapira 2013[35]	Most, but not all, YAs reported having steady emotional support from family and friends. Another important source of support was clinical providers, although a poor clinical alliance was related to increased isolation.	
Lidington 2021[54]	*Facing isolation* emerged as a major theme among YAs, comprised of subthemes including feeling distant from friends, feeling younger than other patients with cancer, and feeling concerned about missing out.	

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TABLE 2 (Continued)

Reference	Results	Meta-inferences
Love 2014[43]	Young men reported feeling unsupported by and isolated from other males.	
Miedema 2007[47]	Some YAs mentioned losing friends since their cancer diagnosis as well as difficulty socializing and making new friends because of decreased energy.	
Mikkelsen 2008[60]	*Needs for social support* emerged as a major theme among YAs after hospital discharge and after completing treatment. Some YAs spoke about a need for connection with same-aged peers with cancer.	
Miller 2015[48]	Many YA women with breast cancer expressed feeling uncertain and socially isolated, especially from same-age peers with no cancer history and older women with breast cancer.	
Milosevic 2020[51]	YAs with breast cancer reported feeling distanced from peers with no cancer history with respect to physical activity and eating patterns.	
Mishra 2018[55]	YAs reported feelings of isolation.	
Musiello 2014[56]	*Isolation* emerged as a key theme among YAs treated in adult hospital settings, primarily related to the age difference relative to older patients with cancer. YAs reported feeling isolated and unable to connect with similarly aged peers going through similar experiences.	
Nizamli 2011[57]	*Social isolation* emerged as a subtheme of *social dysfunction* among female-identifying Syrian YAs with breast cancer. YAs reported that Syrian society has a negative perception of patients who have cancer, associating them with death.	
Panjwani 2019[49]	One third (36%) of YAs spontaneously reported uncertainty related to peers and social life.	
Ruddy 2013[61]	YAs reported that social support decreased during the transition from active treatment to longer term survivorship. Many YAs also wished they could connect with other patients in a similar life stage.	
Siegel 1999[67]	A sense of isolation and of being different from other women was a prominent theme among YAs with breast cancer.	
Snyder and Pearse 2010[44]	YAs with breast cancer expressed feeling isolated in support groups because of a sense that support services and organizations were designed to target the needs of older people with breast cancer.	
Victorson 2019[58]	YAs reported feeling misunderstood, isolated, and alone or incapable of actively engaging in and being open to new relationships because of cancer.	
Mixed-methods studies		
Munoz 2016[46]	Asian/Pacific Islander, Black non-Hispanic, and Hispanic YAs all reported obtaining social support by connecting to friends and strangers using social networking sites, connecting in-person with family and friends, and connecting with others by participating in religious or spiritual activities, such as attending worship services.	• Racial/ethnic minority YAs identified numerous strategies and avenues for maintaining social connection.

Note: The study by Trevino et al.[63] was excluded from this table because the participants were a subset of those from the other trial by Trevino et al.,[39] and all relevant variables were redundant across the two publications.

Abbreviations: MOS SF-36 scale, Medical Outcomes Study 36-item short form; SD, standard deviation; YAs, young adults.

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TABLE 3 Results related to clinical and sociodemographic correlates of social isolation and social connectedness.

Reference	Results	Meta-inferences
Quantitative studies		
Bellizzi 2012[32]	Age was associated with positive cancer-related changes to relationships. Compared to YAs aged 21–29 years, a greater proportion of YAs aged 15–20 years reported more improvements in relationships with siblings (79.7% vs. 61.3%; $p < 0.05$), and YAs aged 30–39 years reported greater improvements in relationships with spouses/ significant others (69.2% vs. 58.8%; $p < .05$).	• Social isolation and social connectedness were related to multiple demographic characteristics, clinical variables, and social determinants of health. • Greater social connectedness was associated with: ○ Increased physical activity; ○ White (vs. non-White) race; ○ Being married/partnered; ○ Breast (vs. other site) cancer; ○ Female sex; ○ Having a stronger alliance with an oncologist; and ○ Being employed or in school. • The relationship of age to social isolation/connectedness was inconsistent across studies.
Brunet 2014[38]	Perceived social support was positively related to physical activity ($r_{pb} = 0.28$; $p < .05$).	
Trevino 2013[63]	White participants (mean ± SD, 42.66 ± 4.44) reported higher levels of total social support as measured by 12 items from the ISEL compared with non-White participants (mean ± SD, 37.50 ± 4.04; $t[62] = -2.26$; $p < .05$). In addition, feeling a sense of belonging or of having someone to engage in activity with was higher among female (mean ± SD, 14.09 ± 2.14) versus male participants (mean ± SD, 12.95 ± 1.98; $t[62] = -2.02$; $p < .05$), White (mean ± SD, 13.88 ± 2.09) versus non-White participants (mean ± SD, 11.50 ± 1.91; $t[62] = -2.22$; $p < .05$), married (mean ± SD, 14.21 ± 1.93) versus nonmarried participants (mean ± SD, 13.04 ± 2.28; $t[62] = -2.21$; $p < .05$), and breast cancer survivors (mean ± SD, 14.59 ± 1.97) versus survivors with other cancer diagnoses (mean ± SD, 13.29 ± 2.12; $t[62] = -2.40$; $p < .05$). Older age was also associated with a greater sense of belonging ($r = 0.29$; $p < .05$).	
Trevino 2013[39]	A greater reported level of appraisal support, or having someone to talk to about problems, was associated with having a stronger alliance with one's oncologist among YAs with advanced cancer ($r = 0.32$; $p = 0.002$).	
Xie 2017[31]	YAs aged 36–39 years (mean ± SD, 43.40 ± 6.36) reported the highest levels of social support, whereas those aged 21–25 years (mean ± SD, 37.21 ± 7.51) reported the lowest levels ($F = 21.74$; $p < .001$).	
Zebrack 2017[64]	YAs aged 30–39 years at diagnosis reported lower levels of social support than those aged 18–29 years at diagnosis ($\beta = -0.27$; $p = .009$), and YAs who were employed or in school reported higher levels of social support than those who were not employed or in school ($\beta = 0.34$; $p = 0.001$).	
Qualitative studies		
Hanghøj 2021[66]	YAs in active treatment and those with impaired immune systems reported particularly elevated loneliness during the first lockdown period of the COVID-19 pandemic.	• Patients qualitatively generated numerous correlates of greater social isolation, including: ○ Cancer type; ○ Being in active treatment; ○ Being immune suppressed; ○ Have cancer-related changes to appearance; ○ Have increased physical late effects of disease; ○ Having a cancer-related genetic mutation; ○ Having young children; ○ Being in medical care settings; and ○ Cultural background. • YAs reported feeling less isolated when they: ○ Had a supportive spouse/romantic partner (YA males only); and ○ Received positive comments or actions from others
Hauken and Larsen 2019[41]	YAs with changes to appearance (e.g., hair loss) and other treatment side effects (e.g., fatigue) reported greater social isolation.	
Hauken 2013[59]	YAs who experienced late effects of cancer expressed increased feelings of social isolation.	
Kenen 2006[42]	Having a genetic mutation was related to increased isolation among YAs with breast cancer carrying a BRCA1 or BRCA2 mutation.	
Knox 2017[52]	Having small children increased feelings of isolation among YAs with advanced cancer.	

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TABLE 3 (Continued)

Reference	Results	Meta-inferences
Lidington 2021[54]	YAs with young children reported feeling particularly isolated and as though they missed out on important developmental changes.	
Love 2014[43]	For YA males, having a spouse/romantic partner who viewed the cancer experience as a joint effort was associated with receiving more effective social support.	
Miller 2015[48]	YAs with breast cancer expressed that social isolation was particularly salient in the context of medical settings and encounters because of the limited presence of other YA survivors.	
Nizamli 2011[57]	Social isolation was exacerbated by a negative societal view of Syrian women with breast cancer; this was related to a cultural understanding that cancer means death.	
Siegel 1999[67]	Many YAs who attended a patient support group reported feeling increased isolation because of a lack of connection with older attendees.	
Zebrack 2010[65]	YA survivors expressed feeling less isolated when they received positive comments or actions from others.	
Mixed-methods studies		
Darabos 2022[68]	YAs with lower levels of perceived isolation preferred to receive support through conversations that were not minimizing and that did not contain expressions of pity/worry (OR, 0.88; 95% CI, 0.78–1.00).	• Social isolation is associated with the type and content of supportive conversations.

Abbreviations: CI, confidence interval; COVID-19, coronavirus disease 2019; ISEL, Interpersonal Support Evaluation List; OR, odds ratio; *r*, correlation coefficient; r_{pb}, point-biserial correlation coefficient; SD, standard deviation; YAs, young adults.

or in school,[64] and have a stronger alliance with their oncologist.[39] In qualitative studies, greater feelings of connectedness were associated with receiving positive comments or actions from others[65] and being a male with a partner who viewed cancer as a joint undertaking.[43] Three quantitative studies explored age at study participation among YAs, with one finding that older YAs demonstrated greater social connectedness than younger YAs,[63] one finding the opposite,[64] and one finding a nonlinear relationship.[31] Specifically, Xie and colleagues[31] reported that YAs aged 21–30 years at the time of participation reported lower social support than YAs younger than 20 years or aged 31–39 years. In addition, one quantitative study explored age at diagnosis. Bellizzi and colleagues[32] reported that a smaller proportion of YAs aged 21–29 years at diagnosis reported cancer-associated improvements in their relationships with siblings relative to YAs younger than 20 years and improvements in their relationships with spouses/significant others relative to YAs aged 30–39 years.[32]

Multiple cancer-related variables were associated with social isolation in qualitative studies. These included greater symptom burden,[41,59] being in active treatment and/or immunosuppressed,[66] having a genetic mutation,[42] and physically being in health care settings.[48] Siegel and colleagues[67] reported that YA breast cancer survivors who attended age-inclusive support groups reported feeling more isolated because of difficulties connecting with older attendees. Nondisease-related correlates also emerged. Two studies[52,54] indicated that having small children was associated with social isolation. Lidington and colleagues[54] elucidated that YAs expressed concern about missing their children's developmental milestones. Culture was also relevant, with Nizamli and colleagues[57] reporting that isolation was exacerbated by the cultural understanding that cancer implies death in their study of Syrian YA breast cancer survivors. Finally, social isolation had implications for communication preferences. Darabos and colleagues[68] indicated that YAs reporting less social isolation preferred receiving support through conversations that avoided minimizing cancer and expressing pity or worry.

Psychological implications of social isolation and social connectedness

The relationships of social isolation and connectedness to psychological well-being were explored in 17 articles (38.6%; Table 4). Within these articles, nine (52.9%) used quantitative methods, and eight (47.1%) used qualitative methods. None used mixed methods. Greater social isolation was generally associated with worse psychological well-being. Brunet and colleagues[38] observed that social support was lower among YAs who reported more stress, and Okamura and colleagues[69] observed that poor social support was associated with greater unmet psychological needs. Conversely, greater social connectedness was generally associated with better

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TABLE 4 Results related to relationships of social isolation and social connectedness to psychological well-being.

Reference	Results	Meta-inferences
Quantitative studies		
Avis 2012[29]	Reported social support scores were lower among YAs diagnosed with breast cancer who also reported more depressive symptomatology compared with those who did not report depressive symptomatology (mean, 4.1 vs. 4.4, respectively; $p < .001$).	• In general, greater social isolation was associated with worse psychological well-being, including: ○ Greater stress; and ○ Greater unmet psychological needs. • Conversely, greater social connectedness was associated with better psychological well-being, including: ○ Less depression; ○ Less anxiety; ○ Less severe grief; ○ Less distress; and ○ Better emotional functioning.
Brunet 2014[38]	Both lower perceived social support ($r = -0.68$; $p < .05$) and not having previously been involved with a support group designed for people with cancer ($r_{pb} = -0.25$; $p < .05$) were associated with greater stress.	
Geue 2019[70]	Positive social support was negatively associated with anxiety and depression at both baseline (anxiety, $\beta = -0.09$ [$p = .008$]; depression, $\beta = -0.10$ [$p = .003$]) and at 12-month follow-up (anxiety, $\beta = -0.09$ [$p = 0.003$]; depression, $\beta = -0.14$ [$p = .001$]). Detrimental interactions were also positively associated with anxiety at baseline only ($\beta = 0.07$; $p = .042$).	
Geue 2019[30]	Positive social support was positively associated ($r = 0.16$; $p < .05$) and detrimental interactions were negatively associated ($r = -0.17$; $p < .05$) with emotional functioning.	
Hoyt 2013[71]	Social connectedness was significantly associated with better cognitive-emotional regulation (r range, 0.67–0.72; $p < .001$ for all).	
Okamura 2021[69]	Poor social support was associated with having greater unmet psychological needs ($\beta = -0.15$; $p = .02$).	
Trevino 2013[63]	Among YAs with advanced cancer, greater overall social support was associated with greater psychological HRQOL ($\beta = 0.43$; $p < .001$), greater existential HRQOL ($\beta = 0.43$; $p < .001$), and less severe grief ($\beta = -0.28$; $p < .05$). Greater reported appraisal support, or having someone to talk to about problems, was also associated with greater psychological HRQOL ($\beta = 0.37$; $p < .01$) and greater existential HRQOL ($\beta = 0.30$; $p < .05$), adjusting for metastatic disease, as well as less severe grief ($\beta = -0.29$; $p < .05$). Greater reported tangible support, or material aid, was associated only with greater psychological ($\beta = 0.47$; $p < .01$) and existential ($\beta = 0.31$; $p < .05$) HRQOL. Finally, greater belonging support, or the sense of having someone with whom to engage in activity, was associated only with greater existential HRQOL ($\beta = 0.34$; $p < .01$).	
Xie 2017[31]	Greater social support was associated with less distress ($r = -0.81$; $p < .001$) as well as less anxiety and depression ($r = -0.84$, $p < .001$).	
Zebrack 2017[64]	Reported social support scores were lower among distressed YAs (mean ± SD, 3.53 ± 0.89; range, 1–5) than among nondistressed YAs (mean ± SD, 4.13 ± 0.78; range, 1–5; $\beta = -0.54$; $p = .001$).	
Qualitative studies		
Easley 2013[45]	One YA with thyroid cancer discussed the psychological effects of mandatory physical isolation after radioactive iodine treatment. They described the treatment as "traumatic" and said that it made them feel like they "had a plague or something."	• Social isolation was related to worse psychological well-being, including feelings of: ○ Distress; ○ Trauma; ○ Anxiety; ○ Depression; and ○ Emotional instability.
Hamid and Khan 2021[36]	Social support from health care professionals was associated with decreased negative feelings, increased positive coping,	

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TABLE 4 (Continued)

Reference	Results	Meta-inferences
	and decreased fears related to disease, treatment, results, and recovery.	• Specifically in the context of the COVID-19 pandemic, for some YAs, increased social isolation enabled time off for recovery, reflection, peace, and quiet. • Increased social connectedness was sometimes related to worse psychological well-being, including feelings of worry and guilt related to feeling responsible for loved ones' emotions, as well as trauma related to hearing other survivors' cancer stories.
Hanghøj 2021[66]	Isolation from others was related to feelings of being alone with negative thoughts, increased anxiety, emotional instability, and symptoms related to depression. Conversely, increased isolation during the COVID-19 pandemic made it easier for some YAs to have time off for reflection and find peace and quiet.	
Kenen 2006[42]	YA breast cancer survivors with strong social support expressed feelings of worry and guilt related to feeling responsible for the emotional well-being of their loved ones. Greater connectedness with other cancer survivors sometimes contributed to "trauma" related to hearing other people's cancer stories.	
Kumar and Schapira 2013[35]	Lack of family and social support was associated with emotional distress and feeling "dejected and frustrated."	
Mishra 2018[55]	One YA explained that being around family and friends was associated with increased feelings of happiness.	
Musiello 2014[56]	Feeling isolated in the treatment environment was associated with feelings of distress.	
Snyder and Pearse 2010[44]	YAs reported that support received from family improved emotional well-being.	

Abbreviations: COVID-19, coronavirus disease 2019; HRQOL, health-related quality of life; r, correlation coefficient; r_{pb}, point-biserial correlation coefficient; SD, standard deviation; YAs, young adults.

psychological well-being. Multiple studies indicated that greater social support was related to fewer symptoms of depression[29,31,70] and anxiety.[31,70] Similarly, Xie and colleagues[31] and Zebrack and colleagues[64] reported that greater support was associated with less distress. Finally, social support was related to improved psychological and existential health-related quality of life,[63] better emotional functioning,[30] and better cognitive–emotional regulation.[71]

Qualitative studies identified psychosocially relevant sources of social connectedness. Hamid and Khan[36] indicated that social support from health care professionals was related to decreased negative affect, increased positive coping, and reduced fear. Kumar and Schapira[35] reported that lack of family support was related to emotional distress. A participant in Mishra and colleagues'[55] study reported feeling happier around family and friends, and Snyder and Pearse[44] reported that family support improved emotional well-being. Specific contexts were also explored. Musiello and colleagues[56] observed that feeling isolated while physically in health care settings was related to distress, and a participant with thyroid cancer in Easley and colleagues'[45] study described physical isolation after radioactive iodine treatments as "traumatic."

Interestingly, Hanghøj and colleagues[66] observed that increased social isolation during the coronavirus disease 2019 (COVID-19) pandemic was related to better psychological well-being because it created time for reflection, peace, and quiet. In addition, Kenen and colleagues[42] reported that increased social connectedness was related to worse psychological health. They reported that YA breast cancer survivors with strong social support expressed worry and guilt about loved ones' emotional well-being and that connectedness with other cancer survivors sometimes contributed to "trauma" from hearing others' stories.

DISCUSSION

This systematic review summarized results from 44 articles reflecting 42 unique samples, most of which were published in the past 10 years. This growing literature highlights the increased attention being paid to YA cancer survivors as an age-defined population and the importance of social isolation and connectedness, broadly conceptualized by Holt-Lunstad and colleagues[7] as the ways in which individuals socially connect, as health determinants. Results across several qualitative studies established the scope of social isolation and connectedness among YA cancer survivors, and descriptive, quantitative studies supported the generation of evidence-based hypotheses. However, only one study was interventional,[64] none used a randomized controlled trial design, and few used validated measures or directly assessed social isolation or connectedness. There is a clear need for future research to fill these gaps.

In several studies, YA cancer survivors reported levels of social isolation and connectedness similar to older survivors and noncancer populations. However, although prevalence may be similar, the risk factors and psychological consequences of social isolation and

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connectedness are likely different for YA cancer survivors given the unique sociodevelopmental context of young adulthood. We identified several risk factors for greater social isolation among YAs (e.g., having young children,[52,54] greater symptom burden,[41,59] being in health care settings[48,67]), and several protective factors (e.g., being married/partnered,[63] female sex,[63] being employed or in school[64]). These factors may identify YA survivors most in need of screening and intervention. Interestingly, the effect of age was not clear. Research specifically designed to test this relationship could elucidate the effect of age within young adulthood on the risk of social isolation. In addition, exploring the effect of developmental stage within the YA age range (e.g., emerging adulthood, young adulthood) also could be important. These life stages can have different developmental goals, which, in turn, could have different implications for the social impact of cancer.

Results also identified important social contexts for minimizing social isolation and increasing connectedness, such as fostering peer connection among YA cancer survivors. In multiple studies, YAs reported difficulty connecting with peer survivors,[44,45,48,52,56,59–61] which sometimes led to increased isolation in environments specifically designed to foster connectedness, such as age-inclusive cancer support groups. Although the need for peer connection has been well established,[72] optimal strategies for facilitating such connection remain unclear. For example, social media is a promising source of social support for YAs,[73] and two included studies indicated that YAs who met other YA cancer survivors through online communities reported lower social isolation and greater connectedness.[53,62] However, recent evidence has also linked high social media use to increased isolation among younger generations.[74,75] There may be a tipping point after which social media stops being helpful, or it may be that the impact of social media varies, depending on the purpose of its use. Future research is needed to explore how best to foster peer-to-peer support and balance online and in-person interactions for YA cancer survivors.

Generally, social isolation was related to worse psychological well-being, whereas social connectedness was often, but not always, related to better psychological well-being. Although there are psychometrically strong measures available to directly assess social isolation (e.g., the Patient-Reported Outcomes Measurement Information System [PROMIS] Social Isolation item bank,[76] the Lubben Social Network Scale[77]) and connectedness (e.g., the Social Connectedness Scale[78]), almost no studies used these measures. Rather, studies used qualitative methods or assessed related constructs (e.g., loneliness, social support) that may not provide a complete picture. Future research is needed that directly assesses social isolation and connectedness using validated measures, in addition to the related constructs already being captured per Holt-Lunstad and colleagues' framework, to yield a more comprehensive understanding of social well-being among YA cancer survivors.

Results of this review may be particularly relevant in the aftermath of the COVID-19 pandemic. During the height of the pandemic, social isolation and loneliness increased for society at large[79–83] and for cancer survivors in particular.[84] Emerging research suggests that the negative social implications of the pandemic may be long-lasting.[85] In one study conducted during the early stages of the pandemic, some YAs reported that increased social isolation enabled time for recovery, reflection, peace, and quiet.[66] It is possible that the adjusted social expectations during this period may have reduced pressure to gather with friends and that the cancer experience was less isolating when more people, regardless of health status, were isolated. However, this may no longer be the case after society's establishment of postpandemic norms. Clinical providers may benefit from assessing social isolation and connectedness as part of YA cancer care in the aftermath of the pandemic.

Limitations

We may have missed eligible articles for this review despite using a comprehensive search strategy. Many included articles did not report participants' ages at diagnosis, and age ranges were often suggested by eligibility criteria. Some articles had age-related eligibility criteria that were broader than those targeted by this review (e.g., aged 15–39 years at diagnosis). In these instances, we considered the average years since diagnosis to determine whether most or all participants were likely within our target age range. For qualitative studies, we only extracted quotes from participants within the target age range. However, some data provided by participants outside our target age range may have been considered. In the future, authors should report both age at diagnosis and age at participation to better distinguish research focused on YA survivors of pediatric cancers from individuals diagnosed during young adulthood. We attempted to identify when multiple articles presented results from a single sample; however, we may have missed instances. Most studies had limited racial/ethnic diversity (except Munoz and colleagues[46]), minimal consideration of other indicators of marginalization (e.g., rurality, sexual orientation, gender identity), and small sample sizes, limiting the generalizability of results. Finally, the heterogeneity of methods made it challenging to compare results across studies or to draw definitive conclusions, and many studies were cross-sectional.

CONCLUSIONS

Social isolation and social connectedness are gaining attention as important health determinants among YA cancer survivors. However, measures to directly assess these constructs are underused, and the impact of interventions on these outcomes remains underexplored. Preliminary qualitative work has established the definition and scope of the problem, and descriptive quantitative exploration supports the generation of evidence-based hypotheses. Research is now needed that assesses social isolation and social connectedness directly, reliably, and validly to support the development and evaluation of translational behavioral interventions.

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AUTHOR CONTRIBUTIONS

Rina S. Fox: Conceptualization, data curation, formal analysis, methodology, supervision, writing–original draft, and writing–review and editing. **Grace E. Armstrong:** Data curation, formal analysis, writing–original draft, and writing–review and editing. **Julia S. Gaumond:** Data curation, formal analysis, and writing–review and editing. **Taylor F. D. Vigoureux:** Data curation, formal analysis, and writing–review and editing. **Corinne H. Miller:** Data curation, methodology, and writing–review and editing. **Stacy D. Sanford:** Conceptualization and writing–review and editing. **John M. Salsman:** Writing–review and editing. **Emmanuel Katsanis:** Writing–review and editing. **Terry A. Badger:** Writing–review and editing. **Damon R. Reed:** Writing–review and editing. **Brian D. Gonzalez:** Writing–review and editing. **Heather S. L. Jim:** Writing–review and editing. **Echo L. Warner:** Writing–review and editing. **David E. Victorson:** Conceptualization and writing–review and editing. **Laura B. Oswald:** Conceptualization, formal analysis, supervision, and writing–review and editing.

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CONFLICT OF INTEREST STATEMENT

Damon R. Reed reports personal fees from SpringWorks and Eisai Inc. for service on Data and Safety Monitoring Boards outside the submitted work. Brian D. Gonzales reports personal fees from SureMed Compliance and Elly Health, Inc., outside the submitted work. Heather S. L. Jim reports grant funding from Kite Pharma and personal fees from RedHill BioPharma, Janssen Scientific Affairs, and Merck outside the submitted work. The remaining authors disclosed no conflicts of interest.

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